

Étienne Mitchell-Bouchard

(514) 606-0203 | etiennemb8@gmail.com | linkedin.com/in/etmibo | github.com/darkzant

PROFESSIONAL OBJECTIVE

Mila Professional Master's student seeking a **required 6-month machine learning credited internship (May–Oct. 2026)** where I can contribute to **applied machine learning systems** in research or industry settings.

EDUCATION

Professional Master's in Machine Learning, Computer Science <i>Mila & Université de Montréal</i>	September 2025 - April 2027 Montréal, QC
Relevant Courses: Deep Learning, Reinforcement Learning, Continuous Learning, Advanced project in machine learning	
Bachelor's in Computer Science <i>Université de Montréal</i>	September 2022 - April 2025 Montréal, QC
- Recipient of the Computer Science Excellence Scholarship from the Ministry of Higher Education of Quebec - Relevant Courses: Machine Learning, Data Science, Introduction to AI	
DEC in Science, Computer Science and Mathematics <i>Collège Lionel-Groulx</i>	August 2020 - May 2022 St-Thérèse, QC

EXPERIENCE

Teaching Assistant <i>Université de Montréal</i>	September 2024 - April 2026 Montréal, QC
Courses given: IFT1119 — Intro to Video Game Programming — Fall 2024 & 2025 IFT2200 — Video Game Programming — Winter 2025 & 2026 IFT1810 — Introduction to Programming — Winter 2025, Winter 2026 IFT2720 — Introduction to Multimedia — Fall 2024 & 2025, Winter 2025 & 2026 IFT1986 — Text Processing Software — Fall 2025, Winter 2025 & 2026 IFT1901 — Introduction to Office Software — Fall 2025 - Graded over 750 student assignments - Helped 350+ students understand course contents - Supervised midterms and final exams in rooms with 100+ students	
Grocery Clerk <i>Maxi Fabreville</i>	July 2020 - January 2025 Laval, QC
- Part-time work to finance my studies (8–16 hours/week) - Use of a forklift, order receiving, shelf stocking, inventory management - Store management, staff supervision, and customer service during closing shifts; securing and locking the store	

TECHNICAL SKILLS

Machine Learning : Python, PyTorch, JAX, Tensorflow, NumPy, Pandas, scikit-learn, SciPy
Tools : Git, Docker, Linux, Google Cloud, Jupyter, Google Collab
Programming : Java, C#, C++, C, JavaScript, Bash, HTML/CSS
Others : SQL, UML, Unity, Data Structures, Algorithms

MACHINE LEARNING PROJECTS

Ilm4mesen Python , ChatGPT & Gemini API , Lua	Paper Github Repo
- LLM-to-game research pipeline to evaluate LLMs on their abilities to play classic 2D games - The LLMs receive optimized multimodal prompts (game screenshots + textual context) to generate controller inputs, which are then directly applied to the game - Benchmarked multiple state-of-the-art LLMs on their reasoning abilities, spatial planning and sequential control, revealing limitations compared to human players	

[rl-continuous](#) | *Python, PyTorch*

- Implemented policy-gradient and Q-learning-based reinforcement learning algorithms for continuous action spaces
- Implemented custom Gymnasium-compatible continuous-control environments to evaluate the algorithms
- Developed a physics-based flash calciner simulator and environment to evaluate RL agents on a realistic industrial control problem with a large state space

[Safe-Autonomous-Driving-MPC](#) | *Python, NumPy, SciPy*

[Github Repo](#)

- Developed a two-stage MPC architecture for safe autonomous driving in dynamic environments, decoupling offline non-linear planning from real-time linearized tracking to balance safety and computational efficiency.
- Validated performance on real-world urban scenarios, achieving high-fidelity tracking with minimal lateral deviation and optimized comfort while demonstrating safe interaction with static and dynamic obstacles.

[AI Price Predictor](#) | *Python, Jupyter, scikit-learn*

- Toy project made in 2020, served as an introduction to machine learning
- Used various machine learning models to predict the direction of a future candle of stocks and cryptocurrencies
- Multiple binary classifiers (MLP, SVM, etc.) were trained on complex calculations derived from past price data

OTHER RELEVANT PROJECTS

[tp-map](#) | *HTML, CSS, JavaScript, Leaflet*

[Github Repo](#)

- Web interactive map that shows players the location of everything that exists in the game
- No interactive map existed for Zelda: Twilight Princess despite high demand following the release of the game's Randomizer, which shuffles the placement of the game's items
- **40-65 daily active users (Google Analytics)**

[SQL-JavaFX](#) | *Java, JavaFX, PostgreSQL*

[Github Repo](#)

- Java application with UI for querying SQL databases
- Used JPA to map SQL tables to Java classes
- Supports custom SQL queries, with basic query templates provided

[Trading Algo](#) | *Python, Binance's API*

- Autonomous cryptocurrency trading algorithm operating 24/7
- Follows a precise algorithmic trading strategy to maximise profit over a long horizon
- Automatically logs executed trades and key market data

[Space Race Unlimited](#) | *Unity, C#*

[Github Repo](#)

- Complete 3D spaceship racing video game
- Supports multiplayer play through LAN with Proton, allowing players to compete and interact with each others
- Contains high score tracking, physics-based movement and spline-constructed courses

[matrix-robot](#) | *HTML, CSS, JavaScript, three.js*

[Github Repo](#)

- Animated robot using only operations on scaling, rotation and translation matrices
- The robot has a walking animation and can look around the scene
- The robot's limbs move independently within realistic joint constraints

[tree-render](#) | *HTML, CSS, JavaScript, three.js*

[Github Repo](#)

- Scene rendering with a tree, leaves, apples, and shadows
- Multiple scenes with tree skeletons to show structure
- Used splines and meshes to create realistic branches

[OnlineShop](#) | *HTML, CSS, JavaScript, jQuery, AJAX*

[Github Repo](#)

- Interactive e-commerce front-end
- Real-time content updates without requiring page reloads
- Full order system that users can utilize